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Preliminary Report of the Paleomagnetism of the Twin Sisters Dunite Intrusion, Washington, U.S.A.

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This letter reports on paleomagnetic and rock magnetic studies of slightly serpentinized (0–15 vol.%) dunite from the Twin Sisters Range and Goat Mountain, northwestern Washington State, U.S.A. The sample areas are restricted to the northwestern and southern margins of the dunite body, but results to date suggest that the remanent magnetism of little-altered dunite in the Twin Sisters Intrusion is stable against alternating field demagnetization and thermal magnetic cleaning, and that it converges to an esatward and down dipping direction with respect to geographic coordinates. The intrusion is believed to be of Tertiary age and, therefore, the large deviation of its direction of remanent magnetization from the usual Tertiary direction would seem to indicate either significant tectonic movement after magnetization or an excursion of the geomagnetic pole into the equatorial Atlantic during Tertiary time. Similar puzzling directions of remanent magnetization have been reported by *Cox* (1957), *Grommé* and *Gluskoter* (1965) and *Saad* (1969a) for Mesozoic and Tertiary igneous rocks of the western United States.

The Twin Sisters Intrusion is located in Whatcom County, Washington, U.S.A., 39 km east of Bellingham and 29 km south of the Canada-U.S.A. border. The main exposure is in the Twin Sisters Range (Fig. 1), where the dunite mass forms an elliptical outcrop pattern 16 km long and an average of 5.5 km wide, elongate to the NNW. A related smaller exposure of dunite is located at Goat Mountain, 7 km SE of the Twin Sisters body. Both dunite exposures are more or less serpentinized on their rims, but the interior of the Twin Sisters body is largely unaltered. Geologic, petrographic, seismic and gravimetric studies have been conducted previously on the Twin Sisters Intrusion (*Ragan*, 1963, 1967; *Thompson*, 1963; *Christensen*, 1971). Gravity studies indicate that the intrusion is only about 1.3 km thick. Samples of Twin Sisters dunite show considerable anisotropy of velocity of longitudinal waves, apparently caused by preferred orientation of the crystallographic *b* axis (lowest velocity) of olivine in the body (*Christensen*, 1971). Paleomagnetic studies are of interest because of their possible bearing on the origin and emplacement of the intrusion, as well as on such problems of regional tectonics as the nature of orogenic deformation in the northern Cascades (*Misch*, 1966) and the possibility of oroclinal deformation affecting large areas in the central North American Cordillera (*Carey*, 1955; *Wise*, 1963; *Watkins*, 1965; *Taubeneck*, 1966; *Greenwood* and *Reid*, 1969).

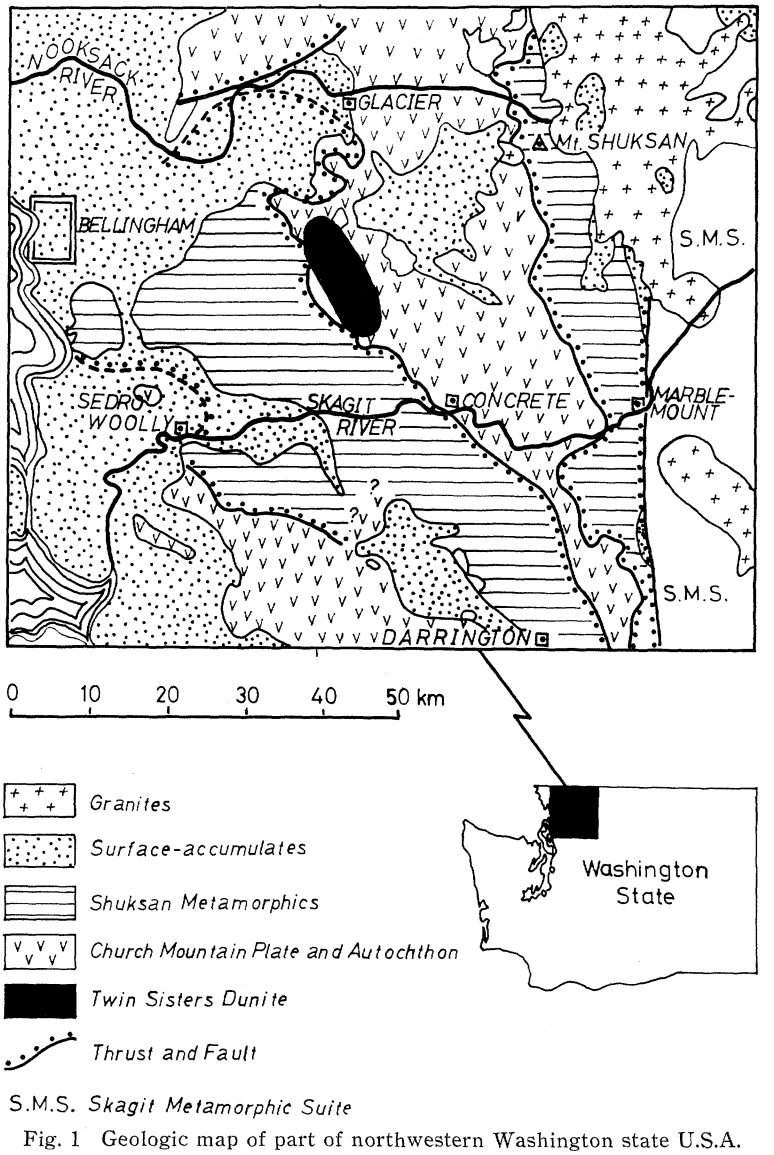


Fig. 1 Geologic map of part of northwestern Washington state U.S.A.

In the present study, nine large oriented blocks of unserpentinized or very weakly serpentinized dunite were chiseled out from the north-western edge of the Twin Sisters body, and a further eight oriented blocks of well—serpentinized dunite were collected from the Goat Mountain outcrop. In both sampling areas samples were spaced over distances of at least a kilometer. Four or five cylindrical laboratory specimens were cut from each block, but because intersample dispersion proved to be extremely low, the sample was treated as the statistical unit in this study. Most stability experiments were performed on groups of specimens which included at least one specimen from each large block. Results of measurement of NRM are shown in Fig. 2 and a statistical sum-

mary is given in Table 1. Alternating field cleaning at a peak field of 200 Oe gave the best convergence in Twin Sisters specimens. All the Goat Mountain samples were found to be unstable (after *a f* demagnetization at 50 Oe their intensities dropped to less than half of the initial value). Results of *a f* demagnetization are shown in Fig. 3 and 4. Thermomagnetic cleaning experiments also were undertaken, and these confirmed the high stability of the Twin Sisters specimens, as shown in Figs. 5 and 6. Best convergence for thermomagnetic cleaning was obtained at 400°C; average direction and statistics after cleaning at this temperature are given in Table 1.

Microscopic, thermomagnetic, and X-ray diffractometer analyses of ferromagnetic minerals from the Twin Sisters and Goat Mountain outcrops reveal the presence of fairly large homogeneous grain of titanomagnetite (less than 10 to a few hundred microns).

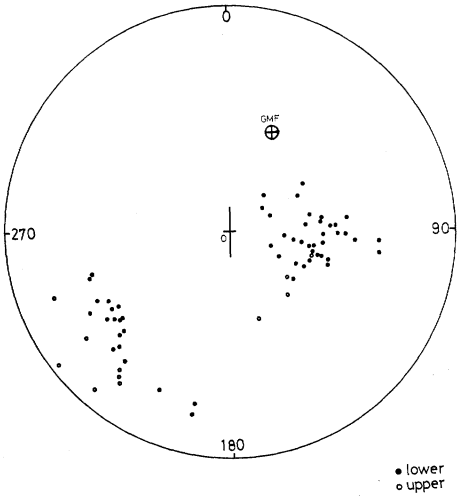


Fig. 2 Schmidt projection of NRM from both Twin Sisters Dunite and Goat mountain outcrops. Hollow circles represent upper hemisphere and full circles lower hemisphere. A cross in a hollow circle show the direction of local geomagnetic field.

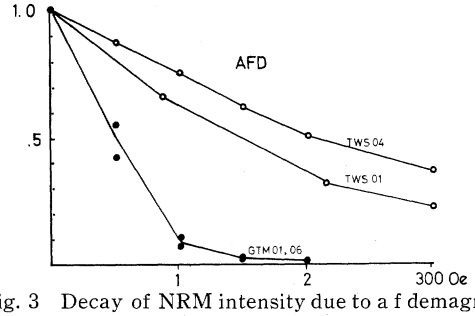


Fig. 3 Decay of NRM intensity due to a f demagnetization. TWS and GTM designate Twin Sisters and Goat mountain, respectively.

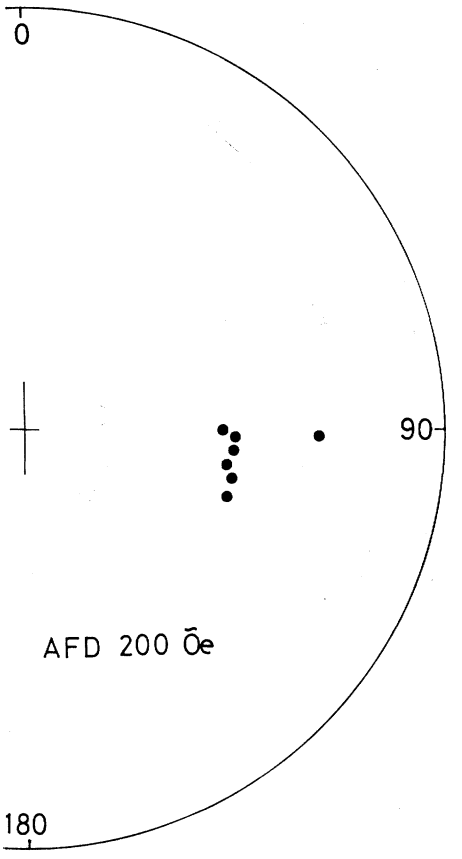


Fig. 4 Dispersion of NRM after a f demagnetization with peak field intensity of 200 Oe projected on Schmidt net. All vectors are downward.

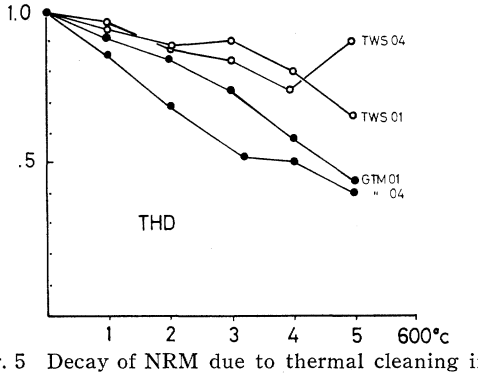


Fig. 5 Decay of NRM due to thermal cleaning in non-magnetic space. Symbols are the same as in Fig. 3.

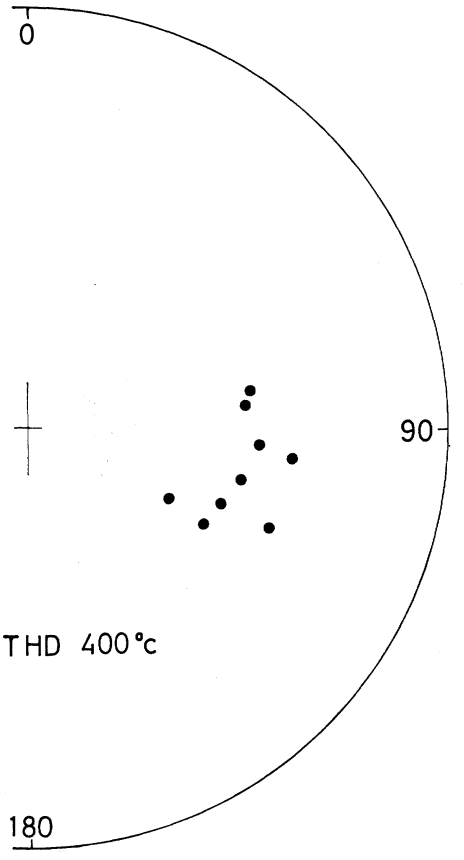


Fig. 6 Dispersion of NRM after thermal cleaning at 400°C in non-magnetic space. All point downward.

The titanium content of these grains probably is very small, judging from the thermomagnetic and X-ray experiments. Grain size distribution of titanomagnetite is poorly studied to date. In addition to titanomagnetite a second potentially ferromagnetic mineral, chromite ($Fe(Fe, Cr)_2O_4$), also is present in these rocks, and it is not certain which opaque mineral, titanomagnetite or chromite, is the most important source of remanent magnetization. Thermomagnetic analyses are shown in Fig. 7 for some ferromagnetics from a sample of dunite and in Fig. 8 for ferromagnetics separated from a chromite vein. As to the effect of serpentinization on the magnetic properties of rocks, several detailed studies have been reported previously (*Cox et al.*, 1964; *Saad*, 1969b). In the present study, the relationships between serpentinization and NRM and between serpentinization and saturation IRM (7.5 kOe field) was found to be ill defined, as shown in Fig. 9. Some of the specimens showed much larger specific gravity than expected of pure dunite of this composition (*i.e.*, $\rho > 3.31$), probably due to the presence of chromite (*Christensen*, private communication), although chromite was not visible on the surface of the specimens in question.

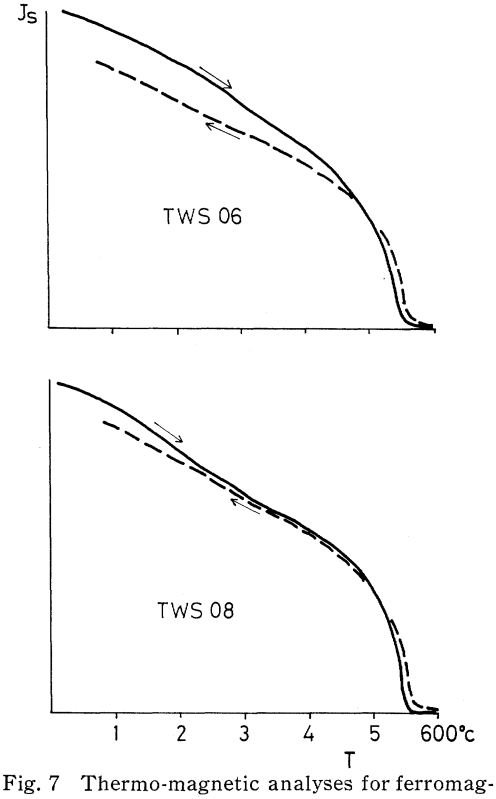


Fig. 7 Thermo-magnetic analyses for ferromagnetic separates of Twin Sisters dunite. Magnetic field of 4.5 kOe was used. Heating in helium gas to prevent oxidation.

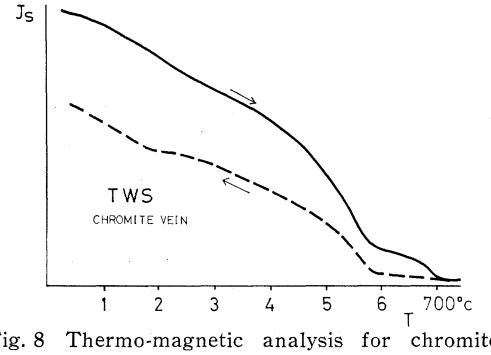


Fig. 8 Thermo-magnetic analysis for chromite minerals from Twin Sisters dunite, heated in air.

Because seismic anisotropy and petrofabric studies described by *Christensen* (1971) have shown the Twin Sisters dunite to possess a strongly anisotropic fabric, it was decided to check for anisotropy of saturation IRM and of *a f* demagnetized (400 Oe) saturation IRM in order to see if magnetic anisotropy could account for the unusual direction of remanent magnetization reported in Table 1. In these experiments a selected

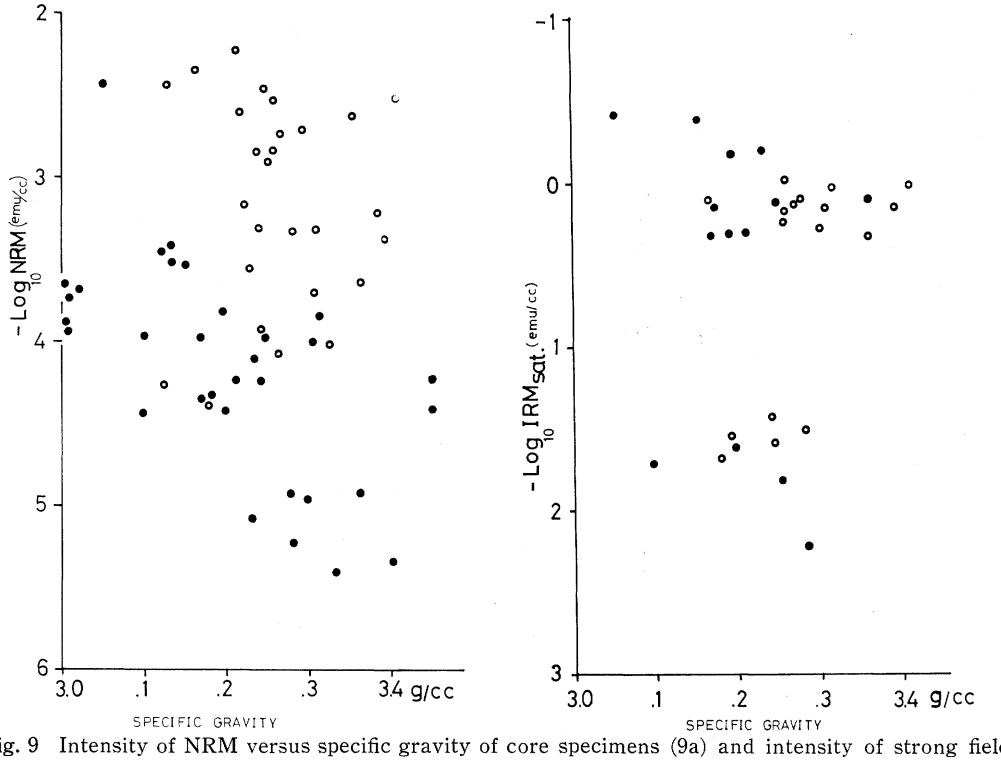


Fig. 9 Intensity of NRM versus specific gravity of core specimens (9a) and intensity of strong field (7.5 kOe) IRM versus specific gravity (9b). Full circles represent Twin Sisters and open circles show Goat mountain specimens, respectively.

specimen was magnetized to saturation every 30° in a plane normal to its axis. A very small degree of magnetic anisotropy was found; deviation of magnetization vectors from the direction of the applied field was not detected, and fluctuations in intensity of remanent magnetization were less than 2.5%. From these observations it seems safe to conclude that there is no relationship between rock fabric and the unusual direction of the stable component of NRM.

On the basis of the present study the authors are convinced that the paleomagnetism of the Twin Sisters Intrusion has considerable potential as a tool for regional tectonic analysis. Further extensive paleomagnetic and rock magnetic studies also can be expected to cast much light on the intrusion mechanics of the dunite body. On the basis of these preliminary studies the authors are planning more extensive work on this problem.

We are very grateful to Professor N. Christensen, Univ. Washington, for his advice and discussion. Some of the specimens used for initial measurements were provided by him. We appreciate also Professor R. Merrill, Univ. Washington, for his support through the experimental work on this study. Suzanne Beske, Paul Johnson, and John Whitney assisted in the field, and their help is gratefully acknowledged. This work was partly supported by NSF grant GA-19233.

Table 1. Paleomagnetism of Twin Sisters Dunite

	N	D	I	α_{95}	k	VPP		d_p	d_m
						Lon.	Lat.		
NRM	9	94.2	60.1	7.2	51.8	54.3	27.1	8.3	10.9
AF Demagnetization (peak 50 Oe)	8	97.7	52.3	7.3	58.4	60.1	19.5	6.9	10.0
AF Demagnetization (peak 200 Oe)	7	99.8	47.7	6.3	92.9	58.5	15.2	5.3	8.2
Thermo-demagnetization (400°C)	9	102.0	49.8	7.6	46.4	61.2	15.3	6.8	10.2

N : number of samples
D : mean declination of NRM vectors in geographic coordinate (degrees)
I : mean inclination of NRM (degrees)
α_{95} : conical angle of 95% confidence of Fisher statistics (degrees)
k : precision parameter of Fisher statistics
VPP: virtual pole position of geomagnetic dipole (degrees)
Lon.: longitude of VPP, westward positive
Lat. : latitude of VPP, north positive
d_m and d_p: elliptical error angle of 95% precision along colatitude direction (d_p) and perpendicular to it (d_m) (degrees)
Locality of Twin Sisters body: 48.6°N, 122.0°W

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